B.Sc. (Hons.) in Environmental Science

B.Sc. (Hons.) in Environmental Science is an undergraduate degree program that focuses on the study of the environment and the various factors that affect it. The program typically takes three years to complete and covers a wide range of topics related to environmental science, including:

1. Ecology: The study of the interactions between organisms and their environment.
2. Environmental Chemistry: The study of the chemical processes that occur in the environment.
3. Environmental Geology: The study of the Earth's physical structure and processes as they relate to the environment.
4. Environmental Biology: The study of the biological processes that occur in the environment.
5. Environmental Physics: The study of the physical processes that occur in the environment.
6. Environmental Policy and Management: The study of the policies and practices used to manage the environment.

The program is designed to provide students with a strong foundation in the natural sciences, as well as an understanding of the social, economic, and political factors that impact the environment. Graduates of the program are well-prepared for careers in environmental science, conservation, and management, as well as for further study in related fields.

Unit 1 - Environment

Definition

The environment is the sum total of all the physical, chemical, and biological factors that surround and influence an organism. It includes both biotic and abiotic components.

Types of Environment

There are two main types of environment: natural and man-made. Natural environments include forests, oceans, mountains, and other natural landscapes. Man-made environments include cities, towns, and other human-made structures.

Components of Environment

The environment is made up of both biotic and abiotic components. Biotic components include all living organisms, such as plants, animals, and microorganisms. Abiotic components include non-living factors, such as air, water, and soil.

Segments of Environment

The environment can be divided into different segments, such as the atmosphere, hydrosphere, lithosphere, and biosphere. The atmosphere is the layer of gases that surrounds the Earth. The hydrosphere includes all the water on the Earth's surface. The lithosphere is the solid outer layer of the Earth. The biosphere includes all the living organisms on the Earth.

Scope and Importance

The environment is important because it provides the conditions necessary for life to exist. It provides us with air to breathe, water to drink, and food to eat. It also provides us with natural resources that we use to build our homes, make our clothes, and power our cars.

Need for Public Awareness

It is important for people to be aware of the environment and the impact that human activities have on it. This can help people make more informed decisions about how they use natural resources and how they interact with the environment.

Ecosystem

Definition

An ecosystem is a community of living and non-living things that interact with each other in a specific environment.

Types of Ecosystem

There are many different types of ecosystems, including forests, grasslands, deserts, and oceans. Each type of ecosystem has its own unique characteristics and is home to different types of plants and animals.

Structure of Ecosystem

An ecosystem is made up of different levels, including producers, consumers, and decomposers. Producers, such as plants, use sunlight to make their own food. Consumers, such as animals, eat other organisms to get the energy they need. Decomposers, such as fungi and bacteria, break down dead organisms and return nutrients to the soil.

Food Chain

A food chain is a series of organisms that are connected by their feeding relationships. In a food chain, energy is transferred from one organism to another as one organism eats another.

Food Web

A food web is a more complex version of a food chain. It shows the many different feeding relationships that exist in an ecosystem.

Ecological Pyramid

An ecological pyramid is a graphical representation of the relationship between different levels of an ecosystem. It shows the amount of energy or biomass at each level.

Balance Ecosystem

A balanced ecosystem is one in which all the different components are in balance. This means that the populations of different species are stable and the ecosystem is able to support all the different organisms that live in it.

Effects of Human Activities on Environment

Human activities, such as food production, shelter, housing, agriculture, industry, mining, transportation, economic and social security, can have a significant impact on the environment. These activities can lead to deforestation, pollution, and loss of biodiversity.

Environmental Impact Assessment

Environmental Impact Assessment (EIA) is a process used to evaluate the potential environmental impacts of a proposed project or development. It is used to help decision-makers determine whether a project should be approved or not.

Sustainable Development

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It involves balancing economic, social, and environmental concerns to ensure that development is sustainable over the long term.

Unit 2 - Natural Resources: Introduction, Classification

Natural resources are materials or substances that occur naturally within the environment and can be used for economic gain. They are classified into two main categories: renewable and non-renewable resources.

Renewable resources are those that can be replenished naturally in a relatively short period of time, such as sunlight, wind, and water. Non-renewable resources, on the other hand, are those that are finite and cannot be replenished, such as fossil fuels and minerals.

Water Resources

Water is a vital resource for life on Earth. It is available from various sources, including rivers, lakes, and groundwater. The quality of water is important for its use, and it is affected by various factors such as pollution and contamination.

Water-borne diseases are illnesses caused by microorganisms present in water, while water-induced diseases are caused by the lack of clean water or sanitation. Fluoride and arsenic are two common contaminants found in drinking water that can cause health problems.

Mineral Resources

Minerals are naturally occurring substances that are extracted from the Earth for their economic value. They are classified into two main categories: metallic and non-metallic minerals.

Material Cycles

Material cycles refer to the movement of elements and compounds through the environment. The carbon, nitrogen, and sulfur cycles are three important biogeochemical cycles that play a crucial role in maintaining the balance of the Earth's ecosystem.

Energy Resources

Energy resources are sources of energy that can be used to generate power. They are classified into two main categories: conventional and non-conventional sources of energy. Conventional sources of energy include fossil fuels, while non-conventional sources include renewable energy sources such as solar and wind power.

Forest Resources

Forests are an important natural resource that provides a range of ecosystem services. However, deforestation and forest degradation have led to the depletion of forest resources, which has a negative impact on the environment and society.

Unit 3 - Pollution and their Effects; Public Health Aspects of Environmental; Water Pollution, Air Pollution, Soil Pollution, Noise Pollution, Solid waste management.

Pollution is the introduction of harmful substances or products into the environment. It is a major problem in America and as well as the world. Pollution not only damages the environment, but it also damages the health of the people. There are several types of pollution, and they are categorized based on the part of the environment which they affect or result which the particular pollution causes. Each of these types has its own distinctive causes and consequences.

Categorized study of pollution helps to understand the basics in more detail and produce protocols for the specific types.

1. **Water Pollution:** Water pollution is the contamination of water bodies, usually as a result of human activities. Water bodies include for example lakes, rivers, oceans, aquifers, and groundwater. Water pollution results when contaminants are introduced into the natural environment. For example, releasing inadequately treated wastewater into natural water bodies can lead to degradation of aquatic ecosystems. In turn, this can lead to public health problems for people living downstream. They may use the same polluted river water for drinking or bathing or irrigation. Water pollution is the leading worldwide cause of death and disease, e.g. due to water-borne diseases.

2. **Air Pollution:** Air pollution is the release of pollutants into the air that is detrimental to human health and the planet as a whole. The Clean Air Act authorizes the U.S. Environmental Protection Agency (EPA) to protect public health by regulating the emissions of these harmful air pollutants. The NRDC has been a leading authority on this law since it was established in 1970.

3. **Soil Pollution:** Soil pollution is the presence of man-made chemicals or other alteration in the natural soil environment. This type of contamination typically arises from the rupture of underground storage tanks, application of pesticides, percolation of contaminated surface water to subsurface strata, oil and fuel dumping, leaching of wastes from landfills or direct discharge of industrial wastes to the soil. The most common chemicals involved are petroleum hydrocarbons, solvents, pesticides, lead, and other heavy metals. The occurrence of this phenomenon is correlated with the degree of industrialization and intensities of chemical usage.

4. **Noise Pollution:** Noise pollution is the disturbing or excessive noise that may harm the activity or balance of human or animal life. The source of most outdoor noise worldwide is mainly caused by machines and transportation systems, motor vehicles, aircraft, and trains. Outdoor noise is summarized by the word environmental noise. Poor urban planning may give rise to noise pollution, side-by-side industrial and residential buildings can result in noise pollution in the residential areas.

5. **Solid Waste Management:** Waste management is the collection, transport, processing, recycling or disposal, and monitoring of waste materials. The term usually relates to materials produced by human activity, and is generally undertaken to reduce their effect on health, the environment, or aesthetics. Waste management is also carried out to recover resources from it. Waste management can involve solid, liquid, gaseous, or radioactive substances, with different methods and fields of expertise for each.

These are the main types of pollution and their effects on the environment and public health. It is important to understand the causes and consequences of each type of pollution in order to take appropriate actions to reduce or prevent its negative effects.

Unit 4 - Current Environmental Issues of Importance

1. **Global Warming**: Global warming is the long-term rise in the average temperature of the Earth's climate system. It is primarily caused by human activities, such as the burning of fossil fuels and deforestation, which release large amounts of greenhouse gases into the atmosphere.

2. **Greenhouse Effect**: The greenhouse effect is the process by which the Earth's atmosphere traps heat and warms the planet. This is a natural process that is essential for life on Earth, but human activities have increased the levels of greenhouse gases in the atmosphere, enhancing the greenhouse effect and causing global warming.

3. **Climate Change**: Climate change refers to long-term changes in the Earth's climate, including changes in temperature, precipitation patterns, and wind patterns. It is caused by both natural factors and human activities, and can have significant impacts on ecosystems, agriculture, and human societies.

4. **Acid Rain**: Acid rain is rain that has been made acidic by certain pollutants in the air, primarily sulfur dioxide and nitrogen oxides. These pollutants are released into the atmosphere by the burning of fossil fuels, and can cause damage to forests, lakes, and buildings.

5. **Ozone Layer Formation and Depletion**: The ozone layer is a layer of ozone molecules in the Earth's stratosphere that protects the planet from harmful ultraviolet radiation from the sun. Ozone depletion refers to the thinning of the ozone layer, which is primarily caused by the release of chlorofluorocarbons (CFCs) and other ozone-depleting substances into the atmosphere.

6. **Population Growth and Automobile Pollution**: The rapid growth of the human population and the increasing use of automobiles have led to increased air pollution, particularly in urban areas. Automobiles release pollutants such as carbon monoxide, nitrogen oxides, and particulate matter into the air, which can have negative impacts on human health and the environment.

7. **Burning of Paddy Straw**: The burning of paddy straw, also known as rice straw, is a common practice in some agricultural regions. This practice releases large amounts of air pollutants, including particulate matter, carbon monoxide, and nitrogen oxides, and can contribute to air pollution and respiratory problems in nearby communities. It is also a significant source of greenhouse gas emissions.

Unit 5 - Environmental Protection

Environmental Protection Act 1986

- The Environmental Protection Act 1986 is an act of the Parliament of India.
- The act is designed to provide for the protection and improvement of the environment.
- The act empowers the central government to take measures to protect and improve the environment.

Initiatives by Non Governmental Organizations (NGO's)

- Non Governmental Organizations (NGOs) play a vital role in environmental protection.
- NGOs work to raise awareness about environmental issues and promote sustainable development.
- Some NGOs focus on specific issues such as wildlife conservation, while others work on broader environmental issues.

Human Population and the Environment

Population growth

- The rapid growth of the human population is a major environmental concern.
- Population growth can lead to increased demand for resources and increased pollution.
- Efforts to control population growth include family planning programs and education.

Environmental Education

- Environmental education is the process of teaching people about the environment and how to protect it.
- Environmental education can take place in formal and informal settings.
- The goal of environmental education is to promote environmental awareness and encourage people to take action to protect the environment.

Women Education

- Women's education is important for environmental protection.
- Educated women are more likely to make informed decisions about family planning and resource use.
- Women's education can also help to promote sustainable development and reduce poverty.